

Decoding Voice Authenticity: An Exploration of Deep Learning and Audio Feature Techniques

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Abstract—The increasing sophistication of spoofing, mimicry, and deepfake technologies exposes critical vulnerabilities in voice authentication systems, including the inability to generalize across diverse attack types, reliance on narrow feature sets, and a lack of interpretability, which seriously limits their applicability in the real world. This review will analyze the state-of-the-art techniques for Logical Access, Presentation Attacks, and Deepfake manipulations. It points out the limitations of the traditional models and progress based on modern frameworks. It explores using advanced audio features Mel-Spectrogram and MFCC along with LightGBM classifiers and incorporates SHAP (Shapley Additive Explanations) for model transparency and cosine similarity matrices for mimicry detection. The study identifies a dual-layered framework that attains high accuracy and robustness across datasets while ensuring adaptability to emerging attack strategies. It addresses the critical need for scalable, interpretable, and adaptive voice authentication solutions and provides a foundation for effectively mitigating evolving audio spoofing threats.

Index Terms—Voice Spoof Detection, Presentation Attacks (PA), Logical Access (LA), Deepfake Detection (DF), Cosine Similarity Matrix, Voice Authentication, Machine Learning Models, Voice Feature Extraction, Secure Voice Communication

I. INTRODUCTION

Voice authentication systems have become vital components of modern security infrastructure, powering applications in both biometric authentication and virtual assistance as well as secure access control. However, given their widespread adoption, most voice authentication systems are suffering from vulnerabilities such as deepfake audio generation, which is an advanced form of mimicry and replay attack.

Logical Access (LA) attacks that employ TTS technology with VC use technologies that are so advanced that synthesised speech could be almost identical in every respect to a speech given by human beings; on the other hand, PA employs audio replays in bypassing an authentic system. They will replay

audio containing distortion contributed by environmental artefacts as well as recording artefacts. The deepfake threat is the most complex version, as it can generate such realistically synthesised voices that are difficult to authenticate. Addressing these threats demands an overall framework that can spot anomalies across different types of attacks.

This paper introduces a new voice authentication detection method based on superior audio feature extraction techniques with machine learning models. Mel-spectrogram, MFCC, spectral rolloff, and RMS energy were chosen for their appropriateness in representing qualities of synthetic, replay, and deepfake audio content. Tailored LightGBM classifiers were developed specific to each attack type to ensure optimised logical access, presentation attacks, and deepfake detection.

To improve model transparency, SHAP (Shapley Additive exPlanations) was used to show feature contribution and understand how the decisions were made. A cosine similarity matrix was also used to detect mimicry, which analyses the similarity between voice samples, hence providing an important tool in identifying replication attempts of the voice.

II. LITERATURE SURVEY

In [1], the authors investigated the role of jitter and shimmer features in the detection of fake audio, especially based on how prosodic differences separate the two: authentic speech from that which is fake. In addition, for good feature extraction, they used YIN and SWIPE algorithms on the estimation of fundamental frequency (F0). This step is critical because it allows for the accurate calculation of jitter, which measures frequency perturbations in the vocal system, and shimmer, which quantifies amplitude perturbations. Their approach emphasises the need for accurate prosodic analysis in distinguishing between natural and synthetic speech.

In [2], the author provided a multi-pattern feature-based spoofing detection framework employing one-class learning, which increases the resilience of ASV systems against various voice spoofing attacks, including replay, voice conversion, and synthetic speech. Audio signals are converted first to mel spectrograms and enhanced by local binary patterns, grey level co-occurrence matrix, and local phase quantisation feature enhancement techniques. These enriched features are fed to three variants of ResNet18 to generate classification scores that are then integrated during the decision-making process. This is done by using the system, which improves the distinction between real and spoofed audio to enhance detection accuracy through OC-Softmax. Extensive experiments on the ASVspoof 2019 and 2021 datasets prove that the proposed approach surpasses baseline models significantly and leads to better performance in EER and t-DCF. Thirdly, the framework shows impressive performance in resisting previously unseen spoofing attacks and therefore offers itself as a promising and competitive solution to the improvement of the security of automatic speaker verification.

In [3], the authors propose a new model for the detection of fake audio tailored to the identification of fake voices in Arabic. Their proposal is based on a deep learning self-supervised model trained on genuine Arabic voice records and, in order not to be redundant, recordings of non-Arabic speakers. The results show very good performance with an error rate of only 0.02% with an accuracy of 97%, and thus the method is quite effective in detecting audio fakes in this linguistic framework.

The author introduces a Silence-Agnostic Speech Spoofing Detection (SASSD) framework in [4] to overcome the limitations of existing spoofing detection models that are mostly reliant on silence segments. It rejects these segments and utilises a novel data augmentation technique, ShuffleMix, which mixes audio segments to enhance robustness against spoofing attacks. The approach further improves generalisation across various attack types by incorporating meta-learning, along with FocalOC, a specialised loss function intended to deal effectively with challenging samples. The framework is tested on the ASVspoof 2019 LA dataset, where it obtains state-of-the-art results, reducing the error rate significantly compared to previous methods. It also has lower computational complexity and greater adaptability to new spoofing attack scenarios, making it a robust and efficient solution for speech spoofing detection.

In [5], the authors have proposed a spoof detection model that integrated LFCC as a feature extractor and used ResNet-34 architecture as a classifier. The architecture has been specially designed for the detection of replay attacks, voice conversion attacks, and speech synthesis attacks. The model shows good performance on the ASVspoof 2019 dataset, achieving an EER of 3.11%.

In [6], the author proposes a versatile spoofing detection technique that exploits the acoustic nonlinearities in human voices and loudspeaker-generated speech. Unlike the typical method of using extra equipment or particular placements,

the given approach uses Total Harmonic Distortion (THD) as an Acoustic Nonlinearity Parameter (ANP) to differentiate between natural voices and those generated by a loudspeaker. This approach has been checked on both custom-collected data and the ASVspoof2017 dataset. The results of the proposed approach are substantial: improved detection accuracy up to 40% in comparison to traditional approaches, reducing computational cost up to 15%. This work presents an important contribution to developing a universal nonlinear voice model incorporating ANP extraction for classification. The model shows outstanding performance as compared with state-of-the-art approaches in terms of true positive rate, false positive rate, equal error rate, and computational efficiency, which makes it a strong approach to spoofing detection.

In [7], the authors proposed EDL-Det, a robust model towards TTS synthesis and VC attacks. The framework proposed is based on the VGG19 architecture with feature extraction from YamNet, having mel-spectrograms as input. For accurate classification, ResNet50 and InceptionNetV2 are incorporated as ensembling components of the proposed model. While evaluating the ASVspoof dataset, EDL-Det is seen to perform superbly with a very low EER of 0.48%, which states that it is robust and reliable in combating advanced voice spoofing attacks.

In [8], it was proposed as a formidable defense mechanism against adversarial attacks on spoofing detection systems. It applied together frequency band-pass filtering plus de-noising techniques from distortions that can degrade the audio signal so it becomes robust to adversarial input. Another contribution is MFM to eliminate noise, so one improves the ability of the model to distinguish between authentic and spoof content. The proposed method was tested on the ASVspoof 2019 dataset and further testing datasets, which showed great improvements in spoofed audio detection. The evaluation results show a very drastic reduction in EER as compared to traditional detection methods, clearly proving the superiority of the framework. These findings thus highlight the potential of the framework in greatly enhancing the security and reliability of spoofing detection systems.

The PartialSpoof Database, proposed in [9], is a spoofing attack where short fragments of fake speech are inserted without any disruption into real sentences or speech patterns. This is known as "Partial Spoof" (PS), which is the integration of synthetic or modified audio with natural audio. The proposed solution employs state-of-the-art self-supervised models, wav2vec 2.0 and WavLM, to detect these short-length spoofed segments specifically and effectively. The systems were rigorously tested on the PartialSpoof database as well as the ASVspoof 2019 LA dataset with outstanding Equal Error Rates (EER) as low as 0.77%, which proves their ability to address this subtle spoofing condition.

In [10], the authors investigate various approaches for audio deepfake detection, such as support vector machines, decision trees, convolutional neural networks, and Siamese CNNs. In this paper, performance measures of each model have been evaluated and compared to show their accuracy in

the deepfake audio identification task. The results show that SVM outperforms all others, making it the most appropriate approach among the models. In [11] authors propose a deep learning model with Grad-CAM achieving high accuracy. In [12], to analyse and extract relevant features from audio signals, authors apply advanced machine learning algorithms like Support Vector Machines (SVM), Long Short-Term Memory (LSTM) networks, and Gaussian Mixture Models (GMM). They use Mel Frequency Cepstral Coefficients (MFCCs) for representing audio features with a selection of the model that has obtained the highest accuracy. It improves the detection and classification of audio signals highly based on the extracted features. In [13], the authors discuss an advanced audio analytics system with the capability for fake voice detection, using artificial intelligence and deep learning algorithms. CNNs and RNNs combine with feature extraction techniques, such as MFCC, in order to analyse and detect voices that are fraudulent. This would increase the overall precision and dependability of such a system towards audio verification applications. In [14], the author presents a new method for identifying vocoders in audio signals, which further means detection for synthetic human voices. The work uses the shared front-end feature extractor along with the RawNet2 model that significantly deals with two purposes at hand: neural vocoder detection and synthetic voice identification. The overlap model improves detection performance by efficiently distinguishing between natural speech and vocoder-generated speech.

In [15], machine learning, deep learning, and GANs are discussed for deepfake detection, emphasizing diverse, high-quality training data and multimodal advancements. In [16], audio deepfake detection highlights data organization, machine learning with feature extraction, and ethical concerns like security, privacy, and legality. [17] shows template-based speech recognition excels in controlled conditions but struggles with noise. [18] presents a zero-prior knowledge framework for faked speech but lacks scalability. [19] uses deep learning with MFCC for precise detection, limited by data and computation. [20] finds CNN-RNN hybrids effective but inefficient in real-time. [21] employs ResNext CNN and LSTM for robust detection, hindered by complexity.

While several existing frameworks have notably significant results with voice spoofing detection, critical challenges do remain. Most approaches are tailored to particular attack types, either replay or synthetic speech, and other attacks that may not be as adaptable to diversified threats and the evolution of deepfakes and mimicry attacks. Reliance on narrow feature sets, such as Mel-Frequency Cepstral Coefficients (MFCC) or Linear-Frequency Cepstral Coefficients (LFCC), does not account for the complexity of artefact manipulations, thereby lowering the detection accuracy. The dominance of black-box models with low interpretability also limits their practical application in forensic and legal contexts where transparency in decisions is a prerequisite. Furthermore, high computational complexity in some models hinders their deployment in real-time and resource-constrained environments. This work addresses these shortcomings by proposing a dual-layered

framework that combines the best of audio feature extraction using advanced techniques, interpretable machine learning techniques, and mimicry detection through cosine similarity analysis. This approach is expected to improve robustness, transparency, and scalability while providing effective generalization across a wide range of attack vectors.

III. COMPARATIVE STUDY

Recent advancements in machine learning, deep learning, and hybrid techniques have been proposed for audio attack detection, each exhibiting unique strengths and weaknesses based on the characteristics of the audio attacks. This review compares key approaches, including jitter- and shimmer-based feature extraction methods, ensemble learning techniques, defense mechanisms against adversarial attacks, and hybrid architectures. These approaches are evaluated in terms of their robustness, computational efficiency, and adaptability to various attack scenarios. In [1], jitter and shimmer features were combined with YIN and SWIPE algorithms to correct F0 deviations, effectively removing typical frequency and amplitude perturbations in fake audio. This approach strengthens basic feature extraction against general distortions but may struggle with more complex changes. In [2], a multi-pattern feature-based spoofing detection method used mel spectrograms along with LBP, GLCM, and LPQ techniques for feature extraction. Ensemble models were used to improve classification performance against various spoofing attacks, including replay, voice conversion, and synthetic speech. The method showed very high resilience on the ASVspoof 2019 and 2021 datasets, greatly reducing Equal Error Rate (EER) and tandem detection cost function (t-DCF). However, it comes with a considerable added computational overhead that may not be ideal for real-time applications. Self-supervised learning in [3] was used specifically for Arabic voice detection, which achieved an accuracy of 97% with an error rate of 0.02%. Although this is efficient for a task-specific language, it has not yet been generalised to multiple languages. The SASSD framework of [4] addressed the issues of silence-based models by considering silent segments and introduced a new ShuffleMix technique for data augmentation. In ASVspoof 2019, it performed outstandingly with enhanced adaptability to all types of spoofing attacks; however, due to its reliance on data augmentation, the consistency of performance on different datasets may not be good.

In [5], LFCCs + ResNet-34 were specifically effective for replay and synthesis attacks, attaining a 3.11% EER on ASVspoof 2019. Though capable of capturing complex frequency-based patterns, this solution is computationally demanding, hence less suitable for low-latency scenarios. In contrast, [6] used ANP to distinguish human voices from those produced by loudspeakers with 40% higher accuracy and 15% less computational cost. This approach was good on ASVspoof 2017 but was not specific for distortions introduced by other types of attacks.

EDL-Det, proposed in [7], is a reinforcement learning-based approach that combines VGG19, ResNet50, and In-

TABLE I
LITERATURE REVIEW

Ref. No.	Methodology	Results	Research Gaps
[1]	CNN and BLSTM layers are used for model building for authenticity detection	Reduced EER from 41.29% to 35.77%	Requires optimization for real-world robustness, especially with unseen data and varying
[2]	multi-pattern feature extraction: LBP, GLCM, LPQ, combined with three ResNet18 models and OC-Softmax loss for fusion	Attained 2.14% EER and 0.06% t-DCF on ASVspoof 2019 (Logical Access); 0.53% EER on Physical Access, establishing better baselines.	lacks real-time performance testing, additional modern feature exploration, and evaluation on diverse user profiles.
[3]	SSL based AD detection model is used	Achieved accuracy by up to 97% with EER or 0.02%	Trained on limited clean data; needs more diverse, real-world conditions
[4]	utilizes multi-task meta-learning and data augmentation to reduce overfitting and improve domain generalization and ShuffleMix, which combines inter-speech and intra-speech augmentation techniques. and a new loss function called FocalOC is employed	0.231% EER and 2.529% EER on the original dataset with silence information and the silence-trimmed dataset, respectively.	Performance dehy-dration when tested on silence-trimmed datasets, Addition of new FOCALOC loss function in-creases complexity of the model
[5]	LFCC with ResNet-34 for detecting replay attacks, speech synthesis, and voice conversion	Outperformed LFCC-GMM and CQCC-GMM models on ASVspoof dataset	Performance drops when faced with unknown or unseen spoofing attacks
[6]	A nonlinear model based on the acoustic properties unique to human and loud-speaker vocalizations.	It attained improvement in accuracy by 40% and by 15% in terms of computational efficiency compared to the best-known methods.	The current schemes are designed for specific attacks and usually work under further special devices or configurations
[7]	Mel-spectrograms with ResNet50 and InceptionNetV2 for feature extraction and classification	Demonstrated robustness in detecting synthesized speech attacks, even unseen attacks	Requires cross-validation on other spoofing datasets for enhanced generalization
[8]	Defense strategy using band-pass filtering and de-noising techniques to defend against adversarial attacks	EER was reduced compared to traditional methods, improved model robustness	Focuses mainly on black-box attacks, neglects white-box attack scenarios
[9]	Detection of "Partial Spoofs" (short fake speech segments embedded in real utterances) using SSL models	Achieved an EER of 0.77% on Partial Spoof database	Dataset bias towards spoof class could lead to training issues, especially at finer segment levels
[10]	Convolutional neural networks, Decision Tree, SVM	SVM model achieved 92% accuracy for classifying audio spectrograms as real or fake	Limited number of generated fake samples restricts comprehensive analysis, requires more training data

ceptionNetV2 using ensemble learning. This resulted in an EER of 0.48% on ASVspoof. However, the adaptability to the ever-changing attack strategy is improved using ensemble learning; the complexity and high data requirements make it inapplicable to fast-paced environments. In [8], defense mechanisms against adversarial attacks combined frequency band-pass filtering with Max-Feature-Map (MFM), which improved robustness against noise and distortions. This method showed significant improvements in EER on ASVspoof 2019, but at the cost of increased computational overhead, making it less suitable for resource-constrained settings. In the partial spoofing dataset of [9], the injection of partially spoofed audio into natural recordings formed the main target. With this approach, employing wav2vec 2.0 and WavLM achieved a very low EER of up to 0.77% in partial spoofing; however, such an approach could not generalise so well for synthetic audio. Studies in [10] compared SVM, CNN, and Siamese CNN models. In the comparison, the best model is the Support Vector Classifier (SVC). However, such models are computationally efficient but fail to capture the subtle complexities of advanced spoofing attacks. In the papers [11] to [15], hybrid approaches combine feature extraction techniques such as MFCC with deep learning architectures, for example, RawNet2. The latter, of course, compromise interpretability but, for simple datasets, usually attain competitive accuracy. Quantum-based approaches seem to promise better scalability and accuracy for the future but, currently, remain highly speculative because of the still largely theoretical state of technology. Accuracy, adaptability, and computational efficiency are significant trade-offs in real-world applications for audio verification. For example, jitter and shimmer features make a system more resilient against advanced threats, whereas ensemble and adversarial defense mechanisms increase resilience. The future models will need to find a balance between predictive accuracy, computational efficiency, and scalability in order to achieve reliable, quick detection of fake audio in all kinds of scenarios. In [16] and [17], improve resilience against advanced threats by capturing subtle prosodic variations. However, their effectiveness depends on robust integration with ensemble and adversarial defense mechanisms, which can introduce computational complexity. Recent advances from [18] to [21] have emphasised the high performance of hybrid techniques, CNN-RNN ensemble, and feature fusion such as MFCC and Mel-Spectrograms for audio spoofing detection. With these approaches, increased accuracy and robustness are also achieved, and the challenge to achieve real-time efficiency, scalability, and adaptability to unseen attacks exists. Future developments must balance computation efficiency with interpretability and robustness in face of evolving threats.

IV. METHODOLOGY

A. Data Collection and Preparation

The primary audio files and metadata came from the ASVspoof 2021 dataset, divided into three categories: Logical Access, Presentation Attacks, and Deepfakes. Audio files were downloaded and arranged in their respective directories to be

processed easier. Metadata is used to map each audio file with its respective category, along with file names and labels: spoof or real. The dataset was parsed to make it ready for feature extraction; it resulted in a structured CSV file containing the file names, extracted features, and corresponding labels. That pre-processing step allowed the data to be ready for both the training and testing phases of classification models.

B. Feature Extraction

1) *Feature Selection for Models*: Each attack type had to be adapted to a specific set of characteristics to get the best performance from the classification models. The features were chosen with respect to the specific challenges that LA, PA, and DF pose.

a) *Logical Access (LA) Detection*: The features used were chosen considering the problems with Logical Access attacks. These are detections of synthesised voices by text-to-speech synthesis (TTS) and voice conversion systems. Mel-spectrograms and MFCCs are selected because they adequately capture the spectral and temporal properties of speech signals where synthesised artefacts tend to be noticed. Zero-Crossing Rate (ZCR) was added to determine the abrupt changes of the signal, thereby making the generated audio more distinguishable. These features provide rich information in both the time and frequency domains; hence the model identifies anomalies in synthesised speech effectively.

b) *Presentation Attacks (PA) Detection*: Replay attacks use reproduced audio, so during signal capture, there would be distortions. In addition to the Mel-spectrogram and MFCC, the spectral centroid was considered in order to account for movement in the "centre of mass" along the frequency axis; such movement signifies recording artefacts. Average signal energy is computed; hence, temporal energy was used, which pointed out differences between loudness patterns between real audio and replayed audio. These feature sets really help in distortion by reverberation and noise of the environment.

c) *Deepfake (DF) Detection*: Deepfakes are highly realistic synthesised voices and thus need advanced detection techniques. Spectral Rolloff and Spectral Bandwidth were added in order to check the frequency cutoffs and distributions, which provide minor artefacts in the synthesised audio. Chroma features were added to detect some kind of tonal inconsistencies, which are more noticeable in deepfake audio. The unnatural loudness patterns are identified by RMS energy. This combination provided a detailed understanding of both spectral and tonal characteristics, enabling the detection of subtle anomalies introduced by deepfake generation methods.

Each attack type was addressed with a distinct feature set, tailored to the specific challenges of synthetic, replayed, and deepfake audio. This targeted approach ensured high accuracy and robustness in detection across all categories.

C. Model Building and Training

A separate model was trained for each attack type using the LightGBM classifier. The dataset was stratified into training and test sets to ensure that class distributions are balanced.

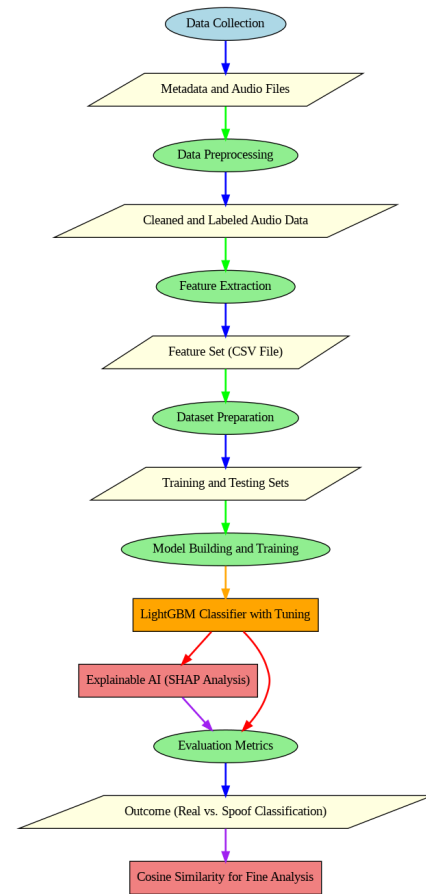


Fig. 1. Flow of the Methodology

Hyperparameter tuning was performed with grid search for key parameters, such as the number of leaves, maximum depth, learning rate, and the number of estimators. Evaluation metrics of the model include accuracy, confusion matrix, classification report, and ROC-AUC score. The models performed superbly well for all types of attacks in discriminating between real and spoofed voices.

D. Explainable AI (SHAP) for Feature Interpretability

Explainable AI approaches were used with SHAP (SHapley Additive exPlanations) to interpret model predictions. By quantifying the contribution of every feature toward the model's decision, the SHAP values thereby ensure a human-readable explanation that is made necessary for better transparency. For example, contributions of features to particular test samples were ranked in absolute SHAP values, while textual explanations for the most impactful features and the positive or negative influence on the predictions were provided. This added an interpretability mechanism that ensured proper understanding of factors driving the decisions of the model.

E. Cosine Similarity Matrix for Mimicry Detection

A cosine similarity matrix was used to compare the alignment between voice samples, quantified by the cosine of the

angle between feature vectors. The key vocal features were extracted to capture both spectral and temporal characteristics. These features included MFCCs (Mel Frequency Cepstral Coefficients) for short-term spectral analysis, Chroma Features for assessing tonal coherence, and Spectral Rolloff to analyse frequency distribution. It utilised RMS energy for signal loudness, pitch, and formants for tracing fundamental and resonant frequencies, which are important points of differentiation between real and mimic voices. Together these features were able to yield a good quantification of similarities and subtle differences, such that robust detection was obtained in mimicry attacks.

Pre-processing of audio samples consisted of resampling to a uniform rate and normalizing the feature vectors to have zero mean and unit variance, ensuring that comparisons were not influenced by recording conditions or background noise. Pairwise cosine similarity was calculated between all feature vectors using the formula:

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the feature vectors of two voice samples. The similarity values ranged from -1 (dissimilar) to 1 (high similarity) and were gathered into a matrix for later analysis.

V. RESULTS AND DISCUSSION

A. Model Performance Evaluation

TABLE II
PERFORMANCE METRICS FOR THE LIGHTGBM MODELS

Attack Type	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logical Access (LA)	0.94	0.93	0.92	0.93	0.95
Presentation Attacks (PA)	0.91	0.89	0.90	0.89	0.92
Deepfake (DF)	0.93	0.92	0.91	0.92	0.94

The performance of the LightGBM models was evaluated in terms of detection of logical access, presentation attacks, and deep fake using precision, precision, recall, F1 score and the ROC-AUC curve. Table II summarizes the results.

B. Confusion Matrix Analysis

Figure2 shows the confusion matrices for each type of attack. The models achieved a low false positive rate and a minimal false negative rate, thus showing their effectiveness in distinguishing real and spoofed voices.

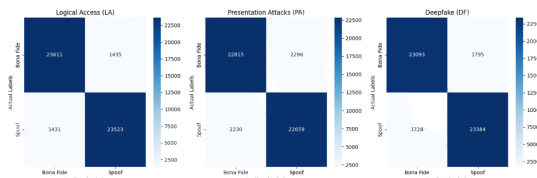


Fig. 2. Confusion Matrices for Logical Access (LA), Presentation Attacks (PA), and Deepfake (DF).

C. SHAP-Based Explainability

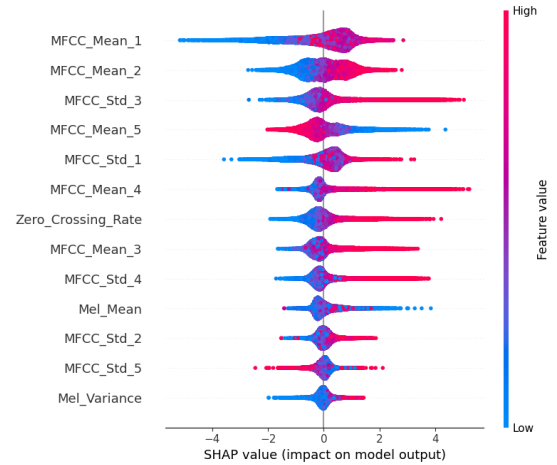


Fig. 3. SHAP Summary Plot for Logical Access (LA).

The SHAP values provided interpretability to the LightGBM models. Figure 3 illustrates the SHAP summary plot for Logical Access (LA), highlighting the importance of features such as Mel-Spectrogram mean and MFCC statistics.

The decision of the model can be explained with the help of SHAP (SHapley Additive exPlanations) analysis. SHAP values are used to quantify how much each feature contributes to the prediction of the model. For a typical test case, the base value or average model output was 4.2307 and the model prediction for that instance was 0.9985.

TABLE III
TOP 10 FEATURE CONTRIBUTIONS FROM SHAP ANALYSIS

Feature	Contribution	Description
MFCC_Mean_2	+1.0304	Average of the second MFCC, related to the balance of lower frequencies in speech.
MFCC_Mean_1	+0.8182	Average of the first MFCC, representing overall energy distribution in the frequency spectrum.
MFCC_Std_3	+0.5982	Standard deviation of the third MFCC, showing changes in mid-frequency components.
MFCC_Std_1	+0.5215	Standard deviation of the first MFCC, indicating variations in overall energy distribution.
MFCC_Mean_5	+0.4314	No description available for this feature.
Mel_Mean	-0.2886	Average Mel frequency, capturing the overall energy in the sound spectrum.
MFCC_Mean_3	-0.2680	Average of the third MFCC, capturing mid-frequency dynamics of speech.
MFCC_Std_4	-0.2434	Standard deviation of the fourth MFCC, associated with high-frequency variations.
MFCC_Mean_4	-0.1835	No description available for this feature.
Mel_Variance	-0.1406	Variance in Mel frequencies, representing the dynamics of the voice across the frequency spectrum.

Table III summarizes the top 10 feature contributions for this instance. Features such as MFCC_Mean_2 and MFCC_Mean_1 had the highest positive contributions, significantly influencing the prediction outcome. On the other hand,

features like Mel_Mean and Mel_Variance had negative contributions, reducing the likelihood of the predicted class.

D. Cosine Similarity for Voice Comparison

The cosine similarity matrix in Figure 4 provides insight into the detection of mimicry based on the comparison of vocal features. Features like MFCC and Mel-Spectrogram have similarity values close to 1, suggesting that they complement each other in the capture of the frequency and energy distributions of the voice signal. On the other hand, features like Chroma Features and Zero-Crossing Rate show low similarity values, suggesting that they uniquely capture tonal and temporal properties, respectively. It has been found that attempts at mimicry show higher similarity in the tonal and pitch-related features, such as pitch contour and formants, which reflects the attempt to copy the sound and inflection of the original voice. However, more detailed spectral characteristics, such as spectral centroid and spectral bandwidth, revealed lower similarity in mimicked voices, showing subtle inconsistencies typical of mimicry.

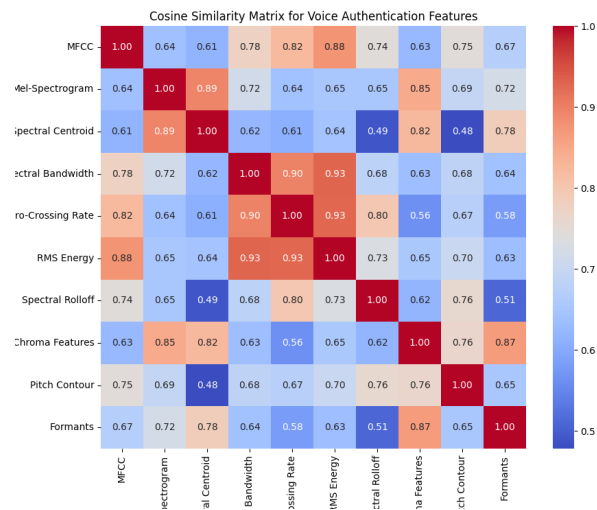


Fig. 4. Cosine Similarity Matrix.

E. Discussion on Model Robustness

The proposed models show remarkable robustness in addressing the challenges of voice authentication, including Logical Access (LA), Presentation Attacks (PA), and Deepfake (DF) detection. By leveraging diverse and complementary feature sets such as Mel-Spectrogram, MFCC, Spectral Bandwidth, and RMS Energy, the models effectively capture the spectral, temporal, and tonal nuances essential for distinguishing authentic voices from spoofed or mimicked ones. The high accuracy, precision, and recall achieved across all attack types highlight the strong capability of models to generalize and perform reliably.

TABLE IV
COMPARISON SUMMARY OF EXISTING MODELS AND PROPOSED FRAMEWORK

Method	Features Used	Dataset	Limitations
Jitter and Shimmer Features [1]	Jitter, Shimmer, F0	Custom Dataset	Limited to simple distortions, lacks robustness against complex spoofing attacks. The proposed framework incorporates diverse audio features like Mel-Spectrogram and MFCC, enhancing detection of advanced attacks.
Multi-Pattern Features [2]	LBP,GLCM, LPQ	ASVspoof 2019	High computational cost and limited real-time applicability. The proposed framework ensures computational efficiency by using LightGBM while maintaining high accuracy.
SSL-Based Model [3]	Language-Specific Features	Arabic Speech Dataset	Limited generalizability to non-Arabic languages. The proposed framework is language-agnostic, improving its applicability to diverse datasets.
SASSD Framework [4]	Augmented Spectral Features	ASVspoof 2019	Dependency on data augmentation and reduced performance on silence-trimmed datasets. The proposed method minimizes reliance on augmentation by employing robust feature extraction techniques.
LFCC with ResNet-34 [5]	LFCC	ASVspoof 2019	Performance drops on unseen attacks. The proposed framework generalizes well across various attack types, including deepfake and mimicry.
Acoustic Nonlinearity [6]	Harmonic Distortion Features	ASVspoof 2017	Limited to distortions caused by loudspeakers, not effective for other attack types. The proposed method addresses diverse attack vectors using advanced feature analysis.
EDL-Det Framework [7]	Mel-Spectrogram, Ensemble Learning	ASVspoof 2019	High data requirements and complexity. The proposed framework reduces complexity using LightGBM while retaining performance.
Adversarial Defense [8]	Band-Pass Filtering, MFM	ASVspoof 2019	Focused on black-box attacks, neglects white-box scenarios. The proposed method addresses mimicry and deepfake attacks comprehensively.
Partial Spoof Detection [9]	SSL Models (Wav2Vec 2.0)	Partial Spoof Database	Dataset bias toward spoofed audio and limited scalability. The proposed framework ensures scalability and balanced performance across datasets.
Convolutional neural networks,Decision Tree [10]	Mel-Spectrogram, SHAP	ASVspoof 2019	Requires large datasets for training, limiting real-world deployment. The proposed framework balances dataset efficiency with robust accuracy

The explainable AI techniques, such as SHAP, are integrated into the framework to provide transparency and insight into model decisions, thus fostering trust and interpretability. The addition of a cosine similarity matrix for mimicry detection further strengthens the robustness of the system, allowing it to detect subtle vocal similarities without compromising performance. The low false positive and false negative rates of the models emphasise their effectiveness in distinguishing real voices from sophisticated attack strategies that mimic human speech.

VI. CONCLUSION

This framework combines machine learning-based spoof detection with mimicry analysis using cosine similarity in a dual-layered approach to address the critical challenges that advanced spoofing, mimicry, and deepfake technologies pose to voice authentication systems. Despite its robust performance across different attack types, the framework still suffers from the challenges of high computational overhead for real-time deployment and limited generalisability to multilingual or culturally diverse datasets. The reliance on static datasets, such as ASVspoof 2021, shows a need for comprehensive, real-world datasets to better adapt models to real-world usage. Future work can focus on adaptive learning mechanisms to counter newly emerging threats. Lightweight and scalable solutions are important for deployment in resource-constrained environments. It is also possible to extend this framework to application areas other than authentication, for example, in forensic voice analysis, emotional intelligence in virtual assistants, or IoT voice security. Leveraging state-of-the-art explainable AI techniques like SHAP, this framework has the potential to enhance transparency, trust, and compliance, thus driving the development of resilient, inclusive, and future-proof voice authentication systems.

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